

Skeleton-Based Action Prediction using Knowledge Transfer

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Abstract

In this work, I extend the GCN-TCN action recognition framework to predict actions given partially observed skeleton-based human motion data. Using the GCN structure and integrating temporal modeling to predict actions, this system aims to accurately predict and classify actions given only partial segments of the whole action being performed. The model aims to use features learned from full videos of actions and apply that knowledge to improve prediction of partially observed actions. This approach has strong performance in accurately predicting actions given limited video/skeleton data with little added computational complexity, and thus has ample potential to be used in real-world scenarios.

1 Introduction

Action recognition is the process of taking video data of someone performing an action and using a combination of their poses and movement to determine what action they are performing. The goal is to turn raw video data into a proper understanding of the action being performed in that video. The ability for computers to autonomously understand human actions being performed via action recognition is a big leap forward in many fields, allowing for far larger quantities of video data to be processed and understood than possible with human interpreters. Action prediction and classification is an extension of recognition; specifically in this paper we predict actions by taking partial videos of actions being performed and classifying them using the knowledge we previously learned from observing full videos. Naturally, this form of action prediction is still applicable for most use cases of action recognition, and is specialized in classifying actions with limited exposure to the action. Therefore, when discussing the applications of action classification, recognition and prediction are both useful for these tasks.

1.1 Applications

1.1.1 Surveillance/Crime Detection

Action classification can be used in real time on surveillance data feeds to classify whether the behavior is suspicious. China already has a vast network of smart surveillance cameras that can detect activities such as fights, theft, or other unusual activity.

1.1.2 Rehabilitation/Healthcare

Action classification can monitor patient movements, track when they fall, and recognize improvement in their movements over time, providing a better rehabilitation experience than what could be achieved



Fully observed actions



Partially observed actions

Figure 1: Partially observed action

alone, without the guidance of a professional.

1.1.3 Robotics

Action recognition will play an important role in the advancing field of human-robot interaction, as it will provide robots with a proper semantic understanding of what humans around them are doing, allowing them to behave in a more sophisticated manner with a deeper understanding of their environment.

1.2 Challenges

1.2.1 Motion Variability

People come in all shapes and sizes. Not only that, people can sometimes perform the same action differently. This level of variability can make it difficult to consistently recognize correct actions. Therefore, it is important to develop a system that can consistently represent human poses (even for people of different shapes and sizes) and handle classification within a certain threshold of variability.

1.2.2 Perspective Differences

People are not always recorded from the same angle, nor are they oriented in the same way in every video. This means that it may be difficult to compare similar actions that are performed from different viewpoints. One way to deal with this is to come up with a way to represent human poses that is independent of perspective, such as the skeleton model (which will be mentioned later).

1.2.3 Body Part Occlusion

No matter how we represent human poses, we still encounter frames in which part of the body is occluded. This can make predicting actions difficult depending on the level of discriminative data that is being occluded. For example, the most discriminative part of the ‘hand waving’ action is the movement of the arm waving side to side, so if only the leg was occluded, we would not struggle to classify the action.

2 Background

2.1 Skeleton Model

The skeleton model is a way to represent a human pose at any given time by representing the joints as nodes on a graph and the bones as edges. This representation has many benefits; first, this representation takes up much less storage than video data, since we only have to store the locations of the joints and which other joints they are connected to, whereas video data must store color values for each pixel. Another benefit is that the skeleton model can easily be rotated and normalized so that all actions are processed at the same orientation, removing unhelpful variation between similar actions. This representation also maintains an intuitive understanding of the human pose, which is still interpretable to us.

2.2 Convolutional Neural Networks (CNNs)

A common deep learning method used for data stored in a grid (such as pixels from images or from frames in a video) is the convolutional neural network (CNN). CNNs apply convolutions to smaller sections of grid data to extract local features, which are eventually used to classify the data. To use these networks



Figure 2: Skeleton model extracted from a frame

for action recognition, we cannot simply use CNNs for each frame of the action (video), since we would fail to capture temporal dynamics. Instead, we could include the ‘time dimension’ to create a 3D array to convolve instead, allowing the CNN (now considered a 3D CNN) to capture temporal dynamics.

2.3 Graph Convolutional Networks (GCNs)

A graph convolutional network (GCN) is a deep learning method that is useful when you can represent data as a graph. For example, the skeleton model represents human poses as a graph where nodes are joints, and bones are edges. This graphical structure can be represented by an adjacency matrix, where each entry A_{ij} is 1 if nodes i and j are connected by an edge and 0 if they aren’t. While training, the adjacency matrix is trained to find the most accurate representation of relationships between joints. This graph structure allows us to model more complicated joint dependency relationships than we could by just using a Euclidean grid. The convolution we use on the graph can be either spectral or spatial. The spectral convolution uses the Laplacian matrix, which can capture better global structure, whereas a spatial convolution aggregates information from neighboring nodes to get a better sense of local information. Since the human skeleton model is a relatively small graph where local features are more important than the global structure, a spatial convolution is typically used for action recognition. The spatial convolution is also far more efficient.

2.4 Temporal Convolutional Networks (TCNs)

A temporal convolutional network (TCN) is a deep learning architecture that uses convolutions to model sequential data. Rather than analyzing individual time steps, TCNs analyze segments of a sequence to determine patterns. The segments that TCNs analyze are not always consecutive frames; sometimes we also use time dilations to look at sections made up of more spread-out frames, so that the TCN can capture short-term and long-term patterns in the data.

2.5 Knowledge Transfer

Knowledge transfer is a technique that helps to classify partially observed actions. Using features obtained from training a model on full actions, we can then transfer knowledge of these features to partial videos. This is a two-stage process; first, we must learn complex features from full actions and a classifier which maps these features to logits for classifying the action. Next, we must find these features in partial videos, and learn a mapping to the embedding space of features from full videos (Cai et al. 2019). Once the partial features have been mapped to the full embedding space, then we can use the classifier learned from full videos to classify the action.

2.6 Additive Margin (AM) Softmax

AM Softmax (Wang et al. 2018) is a loss function which encourages large inter-class distances and small intra-class distances between embeddings. By enforcing a margin between classes, AM Softmax allows the model to learn more distinct features. This is beneficial for knowledge transfer, since it allows the model to maintain class features while also better differentiating between classes.

3 Related Works

Action recognition and prediction have seen several improvements in the last decade using modern deep learning techniques. Many earlier approaches relied heavily on CNNs and LSTMs; these techniques are fundamental and are still actively being used and modified (Kong et al. 2018). In modern approaches, skeleton models are frequently chosen to represent human poses. With the use of skeleton models becoming more popular, graph networks are naturally used to process the adjacency data. (Chen et. al, 2021) proposed a channel-wise topology graph refinement technique for learning spatial relationships, which achieved state-of-the-art results. A modern approach to action prediction was shown by (Cai et al. 2019) who proposed that knowledge from fully observed actions can be transferred to partially observed actions by implementing a learned projection of short video features to the embedding space of full video embeddings. To further separate the distance between inter-class features, the Additive Margin Softmax function was proposed by (Wang et al. 2018), improving classification accuracy. This paper builds upon the GCN-TCN architecture for recognizing fully observed actions proposed by (Chen et. al, 2021) by implementing knowledge transfer techniques proposed by (Cai et al. 2019).

4 Methodology

4.1 Main Model

The first step to this method of action prediction is to build a strong baseline model for classifying fully observed actions. Here I modified a GCN-TCN framework used by (Chen et. al, 2021) which uses several GCN-TCN layers comprised of a GCN and TCN module on each (Shown in Figure 3). The dataset used in testing is the *N-UCLA Multi-view Action 3D Dataset*, so the graph representing connections between joints is initialized as a the known connections for the joints in the N-UCLA dataset. Each dimension (x, y, z) has its own graph to represent the connections between joints, and is updated while training. Over time, the adaptive graph will learn a more relevant adjacency matrix with weighted edges between joints representing their dependence on each other. This baseline model achieves strong results on classifying full action sequences.

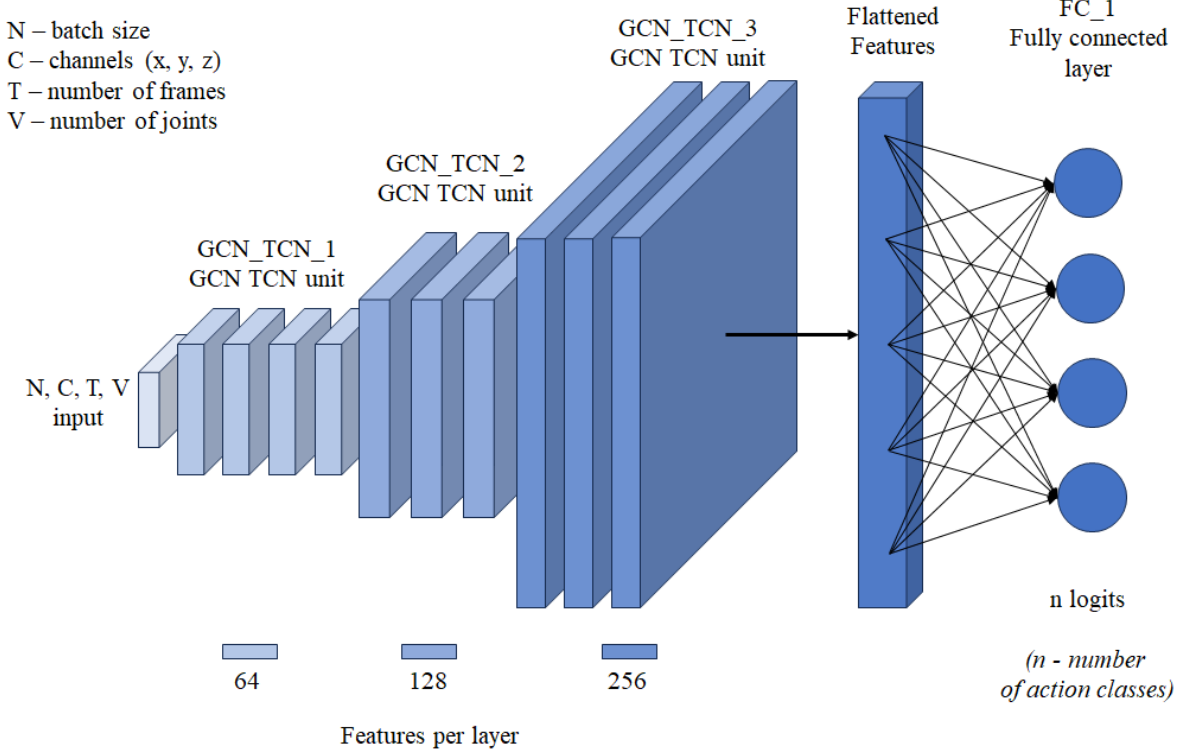


Figure 3: Layers of the main model

4.2 Partially Observed Actions

Now that we have a strong baseline model for predicting full actions, we must decide how to produce partially observed actions. To generate partial actions, we take a cut of the frames from fully observed actions, ensuring that during each training loop we maintain temporal alignment by starting and ending the partial actions on the same frames. These actions are now used as input to train another GCN-TCN model, and we obtain features for partially observed actions. Another potential technique for producing partially observed actions is to take partial actions and linearly interpolate joint data to stretch the length of partially observed actions to the length of the full videos. This way, we could use the full model’s weights to classify partial videos. While this sounds promising, the stretching of partially observed videos leads to a change in the temporal dynamics between frames, and doesn’t produce good accuracy in predicting actions.

4.3 Knowledge Transfer

Once we obtain the features of the partial videos, we train a projection module comprised of two fully-connected linear layers that maps these features onto the embedding space of the full videos. The loss is calculated as the Mean Squared Error between the projected embeddings of the partial videos and the ‘ground truth’ embeddings of full videos produced by the main model. This allows partial videos to benefit from the richer knowledge of the full video embeddings.

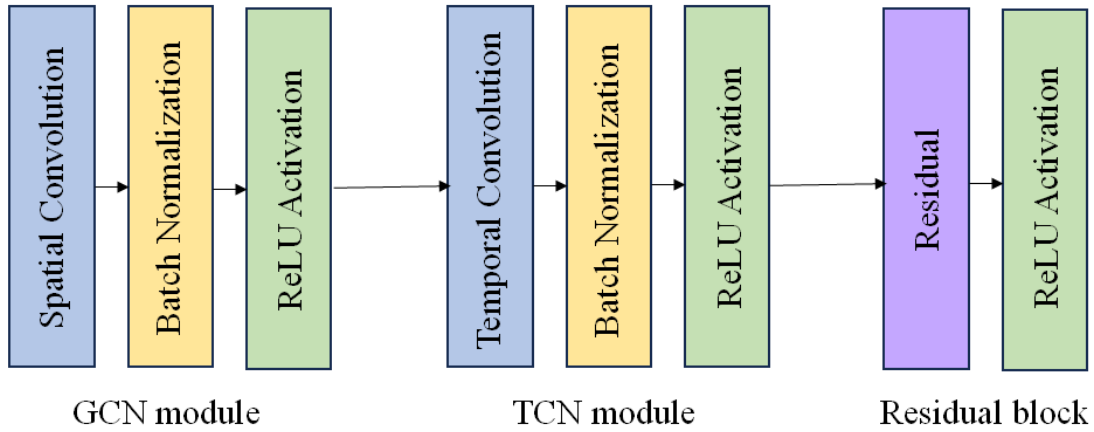


Figure 4: Detailed look at the GCN TCN unit

5 Results

5.1 Discriminative Data Distribution

We can see from Table 1 that there is a significant difference in accuracy between the most and least discriminative portions of the video. This is to be expected, since different actions in the dataset are likely to contain their most discriminative information at different lengths in the video. The largest difference in accuracy is the difference between the middle three segments and the start/end segment. The start and end of the videos usually contain the least discriminative information since they contain less distinguishable movements, whereas the middle portions usually contain the core of the action.

5.2 Knowledge Transfer Comparison

Looking at Table 2, it is important to remark how accurately we can predict partial actions simply by training the main model on the shortened data without transferring any knowledge of full actions. Despite this, we also notice a clear benefit of transferring knowledge of full actions to aid in the prediction of partially observed actions. Compared to the classification accuracy of training on full videos of actions, our methods using partially observed actions keep up relatively well considering that we only use 20 % of the frames.

5.3 Different Video Length Comparison

From looking at Table 3 we observe a noticeable difference in classification accuracy as we increase the duration for which we partially observe an action. Notably there is a large jump from 20% to 50%, after which improvement is slowed. This means that the majority of discriminative information can be found in only 50% of the action. Of course this number would vary between datasets, but it remains an important observation.

Table 1: Top-5 and Top-1 predictions for 20% of the data across five portions of the video.

Metric	Portion 1	Portion 2	Portion 3	Portion 4	Portion 5
Top-5	96.55%	98.23%	98.49%	97.65%	95.88%
Top-1	72.41%	78.21%	81.03%	77.39%	70.24%

Table 2: Comparison of most discriminative 20% using knowledge transfer, same 20% training the model only on the partial data, and full video accuracy.

Metric	Knowledge Transfer	No Transfer	Full Video
Top-5	98.49%	97.12%	99.35%
Top-1	81.03%	74.03%	92.46%

Table 3: Comparison of Top-5 and Top-1 predictions for different lengths of partial videos (all using the most discriminative portion at that length).

Metric	20%	50%	75%	100%
Top-5	98.49%	98.92%	99.28%	99.35%
Top-1	81.03%	87.07%	91.13%	92.46%

6 Conclusion

Using knowledge transfer techniques, we have managed to improve the accuracy of partially observed action classification past simply using a standard action recognition model. The added model complexity to accomplish this is not significant compared to the size of the base GCN-TCN model used in this paper, so there is virtually no trade-off when using this technique to identify partially observed actions. We also notice a moderate difference in which portion of the actions we observed contained the most discriminative information, since many actions that we observed likely contained their most discriminative portions at different moments. In real world situations when we only capture partial actions and want to identify them, we would likely capture different portions of actions on different occasions, and would expect to see a more volatile difference in accuracy between actions. When comparing different lengths of partially observed actions, we see a logarithmic accuracy trend by video length, where once we observe approximately 50% of a video, we seem to have captured most of the discriminative information of the action. This means that in situations where we are likely to be classifying using only partially observed actions, or even purposely using a smaller number of frames to lower computing costs, we can make solid trade-offs and still obtain solid predictions with around 50% of the data. The effectiveness of transferring knowledge gained from full actions to aid in the prediction of partially observed actions is shown in this paper. The applications of accurately predicting partially observed actions are many, given that in the real world use cases (such as surveillance) we don't always end up fully capturing the moments we want to. Given the vast applications and high potential of knowledge transfer techniques, action prediction should continue to see its research grow and innovate. The techniques shown in this paper are already a solid improvement over pure action recognition algorithms for partially observed actions, and should serve as a basis for further innovation.

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